

GRISSOM AFB, IN, USA

WMO: 725335

Lat: 40.650N

Lon: 86.150W

Elev: 812

StdP: 14.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.6	(c) 3.0	(d) -9.3	(e) 3.4	(f) -1.1	(g) -3.0	(h) 4.8	(i) 5.8	(j) 28.9	(k) 26.6	(l) 26.7	(m) 23.4	(n) 9.0	(o) 280	(p) 0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	90.4	75.2	87.9	74.2	85.7	72.9	79.1	85.5	77.4	83.9	75.8	81.8	11.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.7	148.2	81.9	75.6	138.2	80.6	74.0	130.9	79.0	43.2	85.5	41.4	83.9	39.8	81.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.0	(b) 23.0	(c) 19.8	(e) -9.6	(f) 93.9	(g) 7.3	(h) 3.3	(i) -14.8	(j) 96.3	(k) -19.1	(l) 98.2	(m) -23.1	(n) 100.1	(o) -28.4	(p) 102.5		
(4) 26.0	(5) 23.0	(6) 19.8	(7) -9.6	(8) 93.9	(9) 7.3	(10) 3.3	(11) -14.8	(12) 96.3	(13) -19.1	(14) 98.2	(15) -23.1	(16) 100.1	(17) -28.4	(18) 102.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	23.5	28.5	39.7	51.5	63.1	71.4	73.9	72.1	66.2	54.0	41.0	31.3
	DBStd	19.72	12.81	12.04	11.09	9.07	8.81	5.81	5.72	4.81	7.27	9.04	9.83	10.58
	HDD50	2830	824	609	355	89	6	0	0	0	62	299	586	
	HDD65	5876	1288	1023	786	414	146	14	5	72	358	719	1046	
	CDD50	3364	1	6	37	133	412	643	740	686	486	185	30	5
	CDD65	934	0	0	2	8	87	207	280	226	107	17	0	0
	CDH74	7773	0	0	21	81	761	1612	2501	1760	897	138	2	0
	CDH80	2415	0	0	3	6	212	495	885	498	291	25	0	0
	WSAvg	9.3	11.2	10.9	10.5	11.1	9.3	8.4	6.8	6.2	7.1	9.6	10.1	10.5
	PrecAvg	43.8	2.6	2.3	3.0	4.1	4.4	5.2	4.8	4.2	3.8	3.2	3.3	2.9
(15) Precipitation	PrecMax	60.2	7.0	5.7	5.9	10.0	7.7	14.6	14.0	7.9	8.2	7.0	6.6	6.7
	PrecMin	30.2	0.9	0.4	0.7	1.1	1.4	1.1	1.5	1.0	0.9	1.0	0.8	0.8
	PrecStd	7.2	1.3	1.4	1.2	2.0	1.6	2.6	2.6	1.7	1.7	1.7	1.3	1.5
	PrecStd	7.2	1.3	1.4	1.2	2.0	1.6	2.6	2.6	1.7	1.7	1.7	1.3	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	64.0	79.0	80.9	90.0	92.1	95.7	90.3	91.6	83.6	71.5	62.2
		MCWB	55.2	56.0	61.7	63.4	71.2	73.5	78.6	75.5	73.5	68.3	57.6	58.4
	2%	DB	51.4	57.6	68.4	76.1	85.3	88.8	90.9	87.7	87.8	78.1	65.7	57.1
		MCWB	50.0	55.0	59.2	61.8	69.7	73.9	77.3	74.8	73.3	65.2	57.6	55.5
	5%	DB	45.7	51.4	62.2	72.0	82.1	85.9	87.7	85.4	83.7	73.1	60.5	50.7
		MCWB	43.9	49.3	56.1	60.0	68.1	72.6	75.0	73.8	70.8	63.6	55.6	48.9
	10%	DB	39.6	44.9	57.1	67.3	78.4	82.7	84.9	82.7	79.3	69.2	55.4	45.3
		MCWB	38.2	42.8	51.8	57.3	67.2	70.6	74.0	72.5	68.0	61.7	50.1	43.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	60.2	64.5	67.9	76.3	79.5	82.2	80.3	77.3	73.3	63.6	60.0
		MCDB	56.9	61.8	68.2	74.3	81.4	86.5	88.0	85.3	85.5	78.7	68.2	61.8
	2%	WB	50.4	54.6	61.3	64.9	73.0	77.3	79.9	78.1	75.3	69.8	59.9	54.9
		MCDB	51.0	56.8	68.9	71.5	79.1	84.1	86.2	83.8	83.2	73.0	62.5	55.8
	5%	WB	43.7	49.2	57.3	62.5	71.1	75.3	78.1	76.3	73.5	66.4	55.2	49.1
		MCDB	45.1	51.4	61.1	68.4	77.9	80.9	84.7	81.8	80.1	70.5	60.2	50.4
	10%	WB	38.4	42.5	52.4	59.6	69.1	73.6	76.4	74.6	71.5	62.8	51.2	43.9
		MCDB	39.7	44.8	55.8	65.7	75.8	79.6	82.6	79.7	76.1	67.4	55.2	44.9
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	15.5	18.0	20.6	19.9	18.7	18.8	19.3	20.4	19.1	16.4	13.2
		MCDBR	18.6	22.1	24.2	26.0	23.3	21.7	21.0	21.1	24.0	24.1	23.1	19.0
		MCWBR	16.7	18.5	16.5	15.3	12.0	10.3	9.3	10.0	10.1	12.6	15.3	16.7
	5% WB	MCDBR	18.1	21.2	21.2	22.1	19.7	17.8	18.6	17.6	19.1	19.2	20.7	18.0
		MCWBR	17.3	18.8	16.0	16.3	12.4	10.3	9.4	9.8	9.6	13.9	15.9	16.9
(40) Clear-Sky Solar Irradiance	taub		0.299	0.326	0.347	0.410	0.449	0.437	0.452	0.453	0.416	0.358	0.320	0.301
	taud		2.410	2.323	2.359	2.187	2.124	2.208	2.173	2.156	2.240	2.400	2.455	2.439
	Ebn at Noon		273	280	286	273	264	267	262	258	259	266	263	262
	Edh at Noon		26	32	35	45	49	45	46	46	39	30	25	23
(44) All-Sky Solar Radiation	RadAvg		528	803	1146	1488	1707	1938	1937	1724	1415	961	625	439
	RadStd		54	84	90	136	156	127	110	98	113	89	74	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page